

Félix LAPLANTE

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RESEARCH INTERESTS

Causal discovery, optimal transport, conformal prediction, statistical fairness, differential privacy.

EDUCATION

OCT. 2026–PRESENT

PhD in Causality, Université Paris-Saclay (LaMME)

Supervised by Christophe Ambroise, Gilles Blanchard and Pierre Humbert. Latent structure discovery and compatibility constraints in structural causal models.

SEPT. 2024–APR. 2026

Master in Mathematics and AI, Université Paris-Saclay

Ranked 1st, 19.28/20 in M1, 18.21/20 in M2. Joint coursework with the Probability and Statistics program, co-organized with Institut Polytechnique de Paris.

SEPT. 2021–APR. 2024

Bachelor in Mathematics and Applications, Université d'Évry

Ranked 1st, 19.42/20 in L3.

AWARDS / GRANTS

JUNE 2024

Sophie Germain Excellence Scholarship

Also selected for the non-cumulative *MixtAI DATAIA* Excellence Scholarship.

PEER-REVIEWED PUBLICATIONS

DEC. 2024

Scalable Spectral Clustering Based on Vector Quantization

Félix Laplante, Christophe Ambroise

Computo journal paper doi:10.57750/1gr8-bk61.

PREPRINTS UNDER REVIEW

JULY 2026

Contrast-Free ICA and Causal Inference via Wasserstein Distances to the Gaussian

Félix Laplante, Christophe Ambroise, Pierre Humbert

Preprint arXiv:2607.12832, under review at *JMLR*. Submitted for presentation at *ICSDS 2026*.

MAY 2026

A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores

Félix Laplante

Preprint arXiv:2605.25852, under review at an international conference.

OCT. 2025

A General Framework for Joint Multi-State Models

Félix Laplante, Christophe Ambroise

Preprint arXiv:2510.07128, under review at *Statistics and Computing*.

RESEARCH EXPERIENCE

APR.–SEPT. 2026

Research internship in Causality, LaMME

Supervised by Christophe Ambroise and Pierre Humbert. Study of identifiability and causal discovery in structural causal models beyond the Gaussian setting. Development of Wasserstein-based methods for causal discovery.

APR.–SEPT. 2025

Research internship on Joint Models, INRAE

Supervised by Estelle Kuhn and Christophe Ambroise. Development of estimation and prediction methods for joint models and extension to multi-state models. Derivation of uniform high-probability bounds for survival functions, implementation of the *jmstate* package, simulation studies and analysis on real-world data, and comparison with state-of-the-art methods.

APR.–JUNE 2024

Research internship in Clustering, LaMME

Supervised by Christophe Ambroise. Design and implementation of a hierarchical topological clustering algorithm inspired by SVMs, *Spectral Bridges*. Partitioning into Voronoi regions, followed by grouping and automatic detection of boundaries and low-density regions using spectral methods on a local scale-invariant affinity matrix.

TALKS / POSTERS

- JUNE 2026 **A Post-Processing Conformal Prediction Approach for Conditional Coverage via Pivotal Scores**
Poster presentation at *Ukrainian Mathematics Days* (IHP) discussing the PIT-CP framework for conditional validity based on pivotal scores.
- JUNE 2026 **A General Framework for Joint Multi-State Models**
Oral presentation at *Journées de la Statistique* (Clermont-Ferrand) introducing a unified framework for longitudinal biomarkers and multi-state event processes.
- SEPT. 2025 **Uniform High-Probability Bands for Survival Functions**
Poster presentation at *StatMathAppli* (Fréjus) deriving asymptotic uniform confidence bands under a joint modeling framework.
- JUNE 2025 **Selecting the Number of Clusters in a Clustering Algorithm**
Oral presentation at *Journées de la Statistique* (Marseille) introducing a connectivity-based score for scalable spectral clustering.

ATTENDED EVENTS

- MAR. 2026 Workshop on Uncertainty Quantification and Causality (CAUSALI-T-AI, Jussieu)
- OCT. 2025 Winter School on Causality and Explainable AI (SCAI, Jussieu)
- JULY 2025 Working group on Topological Data Analysis (LaMME)
- JUNE 2025 Working group on Causality (MIA Paris-Saclay)

SERVICES

- OCT. 2026 Reviewer for *AISTATS 2027* (upcoming)
- SEPT. 2026 Organizer of LLM sessions at LaMME
- MAY 2026 Co-organizer of a Causality Working Group at LaMME with Pierre Humbert

TEACHING

- 2026–2027 Teaching assistant (TDs), Université d'Évry — 60 hours/year expected

SOFTWARE

- PYTHON **jmstate** — joint multi-state models
pitcp — conformal prediction with pivotal scores
otica — Wasserstein-based independent component analysis
otlingam — Wasserstein-based causal discovery
sbcluster — scalable spectral clustering
fastkmeanspp — C++-backed K-means++
mimisbm — multilayer stochastic block models
uniformbands — uniform bands for survival functions

LANGUAGES

- NATIVE French, English

REFEREES

- PHD SUPERVISORS Christophe Ambroise, Gilles Blanchard, Pierre Humbert
- FORMER TEACHERS Christophe Giraud, Christine Keribin

INTERESTS & ACTIVITIES

- HOBBIES Photography, cycling, piano, hiking